Closing the center-embedding gap in Spanish: can grammatical features do the work of memory?

Norberto Moreno-Quibén[1], Silvia Gumiel-Molina[2], Isabel Pérez-Jiménez[3], and Pilar Pérez-Ocón[4]

[1,2,3,4]Departamento de Filología, Comunicación y Documentación, Universidad de Alcalá (UAH)

Poster Abstract · Cajal Xmas Meeting 2026 · Instituto Cajal, CSIC, Madrid

Abstract

Center embeddings—clauses nested inside one another—push comprehension to its limits, making them a natural probe for the memory architecture that supports language. Their signature is the missing-VP illusion: when one of three required verbs is deleted, readers accept the broken sentence more readily than the intact one, as if a stored prediction had silently decayed. The illusion was recently demonstrated in Spanish (Pañeda & Lago, 2024), with a result mattering beyond linguistics: illusion magnitude tracked the statistical rarity of the verb sequence, while individual working-memory span predicted nothing. Storage capacity, on its own, appears to be the wrong variable.

We ask a question Spanish is uniquely positioned to answer. Spanish offers relativizers that differ in morphological content while occupying identical syntactic positions: featureless *que* versus *el cual* and *quien*, which carry gender, number and animacy. If the illusion arises because degraded memory traces are reconstructed ambiguously, then feature-bearing relativizers should supply distinctive retrieval cues and reduce it—with no change in sentence length, structure, or embedding depth. Featureless *que* should not.

We will test this in 120 speakers, combining speeded acceptability judgements, eye-tracking and EEG, and predict a graded reduction in illusion rate with feature richness, an attenuated sustained anterior negativity during maintenance, and a restored P600 at the offending verb.

The result would place grammatical features, not capacity, at the centre of how the brain holds structure open—and hands a broadly neurobiological question, what makes a memory cue distinctive, a linguistically precise manipulation.

Keywords: sentence processing; cue-based retrieval; working memory; morphosyntactic features; ERP; Spanish

SUPPORTING MATERIALS (NOT PART OF THE ABSTRACT WORD COUNT)

The critical manipulation, on one item. All conditions are double center embeddings of identical depth; the featural content of the relativizer is crossed with grammaticality (VP2 entrevistó present vs. deleted).

(a) La política **a la que** el periodista **al que** los reporteros interrumpieron entrevistó dimitió ayer. [GRAMMATICAL]
*La política **a la que** el periodista **al que** los reporteros interrumpieron ~~entrevistó~~ dimitió ayer. [UNGRAMMATICAL, MISSING VP2]
BASELINE — *que*, gender/number only via the article; the condition used in all prior work.

(b) La política **a la cual** el periodista **al cual** los reporteros interrumpieron entrevistó dimitió ayer. [GRAMMATICAL]
*La política **a la cual** el periodista **al cual** los reporteros interrumpieron ~~entrevistó~~ dimitió ayer. [UNGRAMMATICAL, MISSING VP2]
FEATURE-BEARING — *el cual* inflects for gender and number on the pronoun itself.

(c) La política **a quien** el periodista **a quien** los reporteros interrumpieron entrevistó dimitió ayer. [GRAMMATICAL]
*La política **a quien** el periodista **a quien** los reporteros interrumpieron ~~entrevistó~~ dimitió ayer. [UNGRAMMATICAL, MISSING VP2]
ANIMACY-BEARING — *quien* restricts the antecedent to a human referent.

Differential object marking on *a* blocks a subject-relative reading throughout, so no condition is ambiguous at the relativizer.

REFERENCES

Hahn, M., Futrell, R., Levy, R. P. & Gibson, E. (2022). A resource-rational model of human processing of recursive linguistic structure. *PNAS*, 119(43), e2122602119.

Lewis, R. L. & Vasishth, S. (2005). An activation-based model of sentence processing as skilled memory retrieval. *Cognitive Science*, 29(3), 375–419.

Pañeda, C. & Lago, S. (2024). The missing VP illusion in Spanish: Assessing the role of language statistics and working memory. *Open Mind*, 8.